An IoT-Based Plant Watering System for Controlled Irrigation Based on Plant-Specific Requirements

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Abstract

Water-shortage and low-efficiency irrigation are common problems in the modern world, where about two-third of freshwater are lost in irrigation. The traditional irrigation technique cannot take into account the different needs in water consumption among the plants otherwise over-watering or dehydrating the plant occurs. This inefficiency results in not only wasting of such a fundamental resource, but also reduces plant health and crop yield and negatively impacts soil quality. To address these challenges, the introduction of Internet of Things (IoT) in irrigation systems provides a compelling alternative to the traditional systems that supply water imprecisely to the crops.Over the past few years, the trend of searching for new methods of solutions in agriculture shifted toward automation and smart technologies. Internet of things (IoT) is now highly developed that it can link the controller, cloud services and sensor to develop the system that can perform real time operations. This applies to irrigation whereby one supplies water when the water content of the soil is below a certain level, to make thereof efficient. These systems do more than conserve water; they can also ameliorate the soil condition, increase crop harvests, and shorten the amount of work farmers and gardeners would toil over.

***Keywords*—**IoT, Smart Irrigation, Soil Moisture, Water Conservation, Plant-Specific Irrigation

Introduction

Water is one of the most precious resources on the earth and needs to be carefully managed. Agriculture indeed dominates water use in Brazil and consumes almost 70% of the world's total freshwater. According to the Food and Agriculture Organization (FAO), Population and urbanization growth and climate change are causing the water supply-demand gap to increase rapidly. Water scarcity is a vital environmental issue and for many areas food security along with sustainable development depends directly on water supply. The wide spreading of the irrigation techniques without consent, on the other hand, contributes towards considerable water waste, which could have been conserved and more efficiently used.

Conventional irrigation methods flooding, sprinkler-based systems, or manual watering are on a rigid schedule. and don't reflect the true soil moisture conditions or plant requirements.

Yet in private gardens, cities and fields it's customary to distribute the same amount of water to all plants no matter what kind. But water needs for the species vary greatly: the aloe vera, for instance, needs very little water, but an ornamental rose is said to require frequent watering. Consequent to this uniform .

To curb these inefficiencies, scientists and engineers are considering turning to smart irrigation (M. Panday et al.(2022))systems which resort to automation and sensing technologies. (P. Meena et al.(2023)), The Internet of Things (IoT) (A. Singh et al.(2019)) has been of particular interest during the past few years. IoT can be used to collect data in real-time (P. Roy et al. (2021)), perform remote monitoring and make intelligent decisions, connecting sensors, microcontrollers and communication modules. When used in irrigation, IoT(S. Banerjee et al. (2021)) enables farmers and household customers to check soil moisture, temperature and humidity and then provide an appropriate amount of water at the time when it is required. Not only does this type of precision agriculture minimize wastages of water, it also enhances the health of the plants, it saves labor, and it maximizes productivity.

Even though a number of irrigation systems utilizing the IoT have been developed, most of them are created to manage water feeding depending on soil moisture indications or environmental alterations. Although this method will save water when compared to manual irrigation, the method fails to consider the fact that various plants in the same garden or farm will need varying quantities of water. What is the optimal moisture level of one crop may not be adequate or may be too much in another crop. This study recognizes this gap and suggests a plant-specific water irrigation solution which employs the use of IoT technology to provide a tailored amount of water to each plant.

It incorporates soil moisture sensors, solenoid valves, water pumps, and microcontrollers, including the Arduino and ESP32. The threshold value is established on each plant according to the requirement of water. Once the moisture content of the soil containing a given plant is below the setpoint, the controller will then open the solenoid valve to supply the exact amount of water. Flow sensors are integrated to determine the specific volume being delivered.

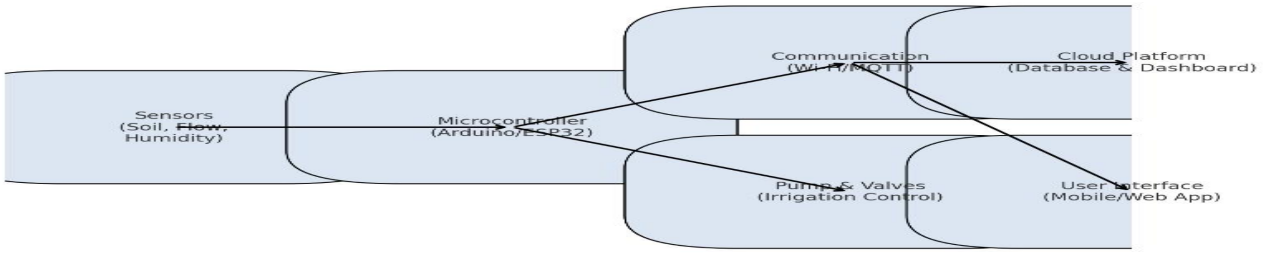


Fig. 1. System Architecture of IoT-Based Plant Watering System (Source: S. Mishra et al.(2020),)

Literature review

In the last 20 years, the issue of water shortage has caused scholars to seek effective approaches of irrigation(N.Gupta et al.(2019)), (A.Singh et al.(2022)) that will reduce wastage and enhance the productivity of crops. Initial automated irrigation was mainly on threshold-driven soil moisture systems, whereby a sensor was used to measure the soil conditions and a pump was turned on when the moisture level fell below a predetermined point. An example is Smith and Patel in (R.Kumar et al.(2018)), who found that this method was effective in minimizing the use of water as opposed to manual watering. Yet, their system used the same threshold to all crops and this failed to reflect the differing physiological requirements of different plant species. This constraint was evident as water sensitive crops became over-irrigated whereas drought resistant ones were not over-irrigated. On the emergence of the Internet of Things (IoT) resulted in the implementation (M. Kumar et al.(2022)) of communication modules, cloud connectivity, and mobile-based interfaces to irrigation systems. (M. El Mouatamid et al.(2020)), According to (P.Patel et al.(2020)), they developed a GSM-based model through which farmers are able to get SMS notification of soil moisture status and remotely operate pumps. Their work became a great leap forward because it also had the element of real time feedback and remote access. However, the model still relied on manual input on the part of the user, which restrained its independence. Much later Kumar et al. presented Wi-Fi-enabled controllers that were connected to cloud dashboards that continuously recorded soil and climate data. The farmers might see how it was and how it could be with more or less moisture and set the watering hours corresponding and Although this help in improving the level of moisture control and decision-making, it still use homogenous soil moisture thresholds in the sense that the needs of plants were irrigated in the same way.

In line with the establishment of IoT-related systems, the idea of precision agriculture has arisen(M.Thomas et al.(2021), S. Patel et al.(2023)), aiming at distributing water, fertilizers, and other resources precision-based fashion used machine-learning-based IoT (R. Patel et al.(2023)), sensors to forecast the irrigation needs of wheat fields and achieved a 30 percent decrease in water consumption. In spite of such an effective predictive approach that saved resources, the approach was developed in large-scale monoculture farming where the demand of water is more or less constant across different crops. adopted a vineyard irrigation system, which was an IoT implementation and took into account the microclimatic variations in different parts of a field. Their model optimized irrigation displacement according to changes in soil and environmental information and revealed how IoT might have a positive impact on decision-making on a local level. Yet, even this system was not created to work in gardens or on the farms, where there may be numerous plant species with different water needs.

Another field of research interest has been the cloud integration. showed how Google Firebase can be used as a back-end to an irrigation model and allow data storage and visualization with a mobile app and make decisions. On the same note, ThingSpeak was used to process real-time sensor data and display the soil conditions in a graphical form. These platforms gave better access to the farmers and minimized the use of measurements which were manual. Although these advantages were achieved, the logic of irrigation was once more generalized and there was no care to set various thresholds or amount of water to be used in the particular plant species.

The current study is based on this gap. In contrast to previous methods, the system proposed incorporates IoT sensors and microcontrollers to provide each plant with various soil moisture thresholds. Irrigate each plant when the threshold is crossed, so that roses, tulsi, and succulents though planted the same soil are watered separately and at the right time. Using this method fills the gap between the large-scale precision agriculture and the small-scale mixed plantations and provides a highly adaptable and scalable solution. Through individual plant-level water distribution, the system can minimize the wastage of water in addition to yielding healthier growth, which is beneficial both ecologically and economically.

Methodology

A. System Architecture

The IoT-based plant watering system intended to be proposed is to automatize irrigation based on soil monitoring (V.Sharma et al.(2021)), and provide water based on the specific needs of various plants. The architecture is divided into three big layers, sensing, processing and actuation. Sensing layer It is provided with soil moisture sensors installed at pots or beds to monitor the soil moisture. The processing layer consists of a microcontroller (Arduino UNO or NodeMCU) (A. Kumar et al.(2022)) to which sensor readings are

Used in system when thresholds are applied. Miniature water pumps and solenoid valves within the actuation layer are activated to supply a particular amount of water to any given plant.

B. Hardware Components

The system utilizes easy to obtain and affordable components:

- Soil Moisture Sensor: This is used to measure the amount of volumetric water in soil based on a change in resistance or capacitance.
- Microcontroller (Arduino): Acts as the control unit to process sensor data and send/receive information to the IoT platform.
- Relay Module: Intended to open and close pumps or valves in a safe way.
- Mini Water Pump: This guarantees that water is sent to the plants accurately.
- Power Supply (Battery): Powers the microcontroller and sensors.
- Wi-Fi Module (inbuilt ESP32): Enables IoT connectivity for remote access.

C. PreproceSoftware Design

The software layer of the proposed system is critical in relating the sensing hardware to the user via the IoT platform. The microcontroller is programmed in such a manner that it constantly gathers data in the form of soil moisture sensors and compares it to set threshold values. An algorithmic method of simple decision-making is applied, in which every plant has an optimum range of moisture based on its water needs. The controller is activated by a sensed value which is below the set point causing the relay to make contact with the pump or solenoid valve. This watering procedure continues at a set amount of time in order to deliver the plant with only the necessary amount of water. When the soil is sprinkled to the ideal moisture content then the pump is automatically switched off to avoid over-irrigation.reprocessing

The controller is connected to an IoT service via Wi-Fi to an IoT platform like Blynk or ThingSpeak to offer remote access and monitoring. (K. Soni et al.(2022)), This is what facilitates transmission of data in real time, which is stored on the cloud and shown on a web or mobile dashboard. The user is able to view the trends of soil moisture, monitor the use of water, and get a notification of the irrigation processes. Moreover, the system has a manual override option whereby the user can remotely turn on or off watering in case it is necessary. The software is designed to keep the process of irrigation intelligent and friendly through this integration of sensing, control logic, and IOT connectivity

D. Working Principle

The general operation of the system can be considered the closed-loop control principle in which sensor data is watchedand contrasted with fixed parameters constantly. All the plants are subjected to an appropriate water content in soil by its category thus ensuring that it is the plant itself, which points to the water supply rather than having a homogeneous supply of the water. When the moisture reading of the soil falls below the threshold, the microcontroller will actuate the actuation module as well as turn the water pump or valve that is attached to it on. The pump operates over a set period of time, which is pre adjusted hence guaranteeing the plant a specific level of water quantity. Upon irrigation, again the sensor measures the new soils condition and the system determines whether more water is necessary or purposely off the pump.

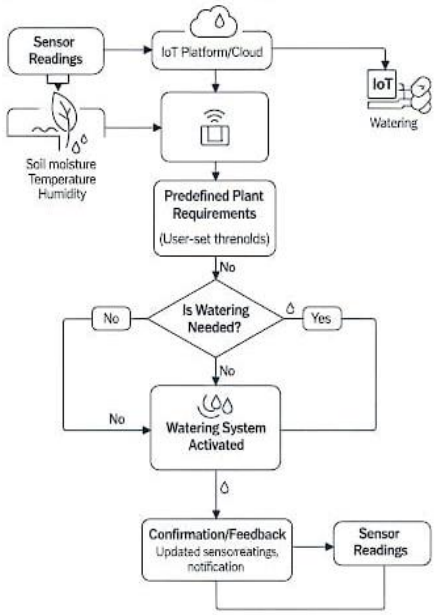


Fig. 2. Flowchart of system working process

This process is automated and operates without the need of human intervention, though the added IoT connectivity claims that a user is never ignored on the process. The user can access real-time sensor monitoring, water consumption history, and go on or off water with a mobile application or web interface. Alerts will also be produced every time a watering operation is triggered and when it has been finished. The system can also be put into manual operation in the event of network failure or user preference where remote start and shutdown of irrigation could be done. In this working principle, automation, accuracy, and remote monitoring are briefly integrated to form a sustainable and efficient irrigation system (R. Kumar et al.(2018)) which conserves water even though the plants would remain healthy. It operates on a basic feedback mechanism of air feedback as the soil sensor constantly checks the level of moisture around the roots of the plant. The results of such readings are calculated against a set of predetermined threshold values determined based on the kind of plant. When the soil is detected to be dry the microcontroller switches on the water-pump via a relay then when the moisture level is detected to be enough the pump is turned off. This cycle makes every plant get the correct quantity of water.

E. System Implementation

The proposed system consists of the combination of hardware, software, and IoT. Moisture sensors were also presented on the soil of both the plants to acquire real time measurements. These were connected to a microcontroller (NodeMCU/Arduino) that computing power, used to interpret the readings and compare it to predefined thresholds imprinted in its memory. We put them on a calibration of different types of plants in such a way that its thresholds aligned precisely to their water requirements.

And we then wired the microcontroller to a relay-driven water pump pumping water into each of the plants out of a reservoir. To all the plants a small pipe or drip spout was assigned, and by this means we could spread out the water accurately. In the entire design, an IoT-based platform was connected to Wi -Fi and allowed us to have remote control and real-time monitoring. A mobile dashboard gave us soil moisture trends, watering records, and usage statistics, and and push notifications were used to notify ourselves when irrigation was turned on.

In order to test the efficacy of the design we simulated some scenarios once using different groups of plants, but artificially altering the soil moisture in order to simulate dry and wet conditions. The system worked accordingly that is, when a pump was fitted, the pump was switched on and off according to the needs of each plant. This tested the concept and we then deployed physically. It was configured as the soil moisture sensors (R. Kumar and et al. (2021)), were connected to a microcontroller which served as the central controller. I put the sensors directly next to the roots to detect real-time sensor values of the soil.

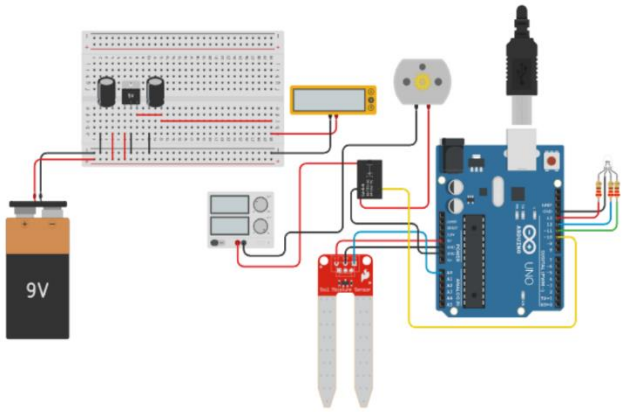


Fig. 3. Conceptual prototype of smart watering setup.

Results

Plant watering was planned as the IoT-centered system that should optimize the irrigation system with respect to the soil moisture level and the individual needs of plants. The system has also been tested through a conceptual test to evaluate its efficacy and in comparison with traditional manual watering process. The results indicate palpable variations of both the soil moisture control and the total average water use. Results are reported in the form of the stability of soil moisture, irrigation efficiency and relative water savings of the various plant types. Among the most valuable study observations is the difference between the resolution of the soil moisture of the IoT-controlled irrigation and manual watering. When it came to the manual watering example, the content of the soil moisture was more likely to be affected radically. As water was used periodically and not depending on the real state of the soil, it tend to cause a process of over watering then dryness. The trend not only consumed excess water, but also resulted in poor soil because roots were either submerged in surplus water, or would be repeatedly underwater only to face the challenge of too little water. This volatility causes the problem in keeping the plants in a manner that ensures constant growth as even delicate varieties that are vulnerable to water stresses find a hard time with them.

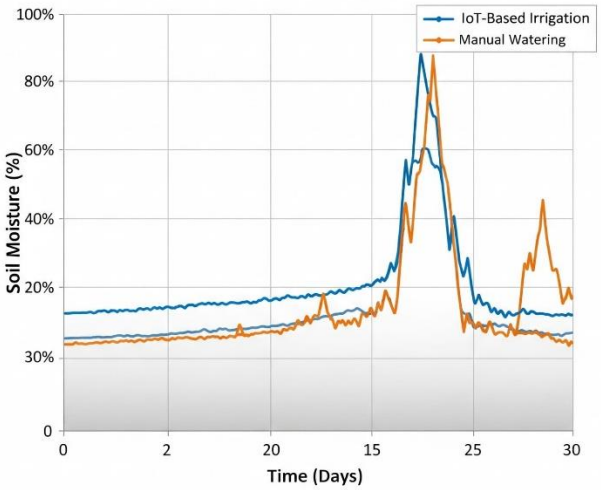


Fig. 4. Soil moisture vs. time comparing IoT-based irrigation and manual watering.

Compared to the IoT-based irrigation system, Figure 4 reveals that soil humidity is observed to be lower after 30 days of established under manual water application to the ground. One can obviously see how manual irrigation leads to great variability where the soil moisture depletes too low or becomes excessive after overwatering. These drastic changes happen due to the fact that use of water was not regular or controlled. Such unpredictable observations have the capabilities of stressing plants, which then undergo strain in lengthy waterlog phases, which are then succeeded by dry proliferation. The system based on the IoT, in contrast, has a much more stable curve. Despite some variations that exist, the percentage of soil moisture stays at a desirable level during the period of observation. This is because the pump will be automatically triggered when the soils in the soil become drier than the required moisture. A steeper increase and decrease in the IoT curve than in the case of the manual counterpart can be observed even in peak watering instances where the IoT approach is better with regards to taking control over the irrigation processes. This is important in terms of stability as far as the health of the plant is concerned. This can be ensured by making sure the soil is consistently moist so that roots do not tend to be subjected to more normal situations. It also reduces waste of water since irrigation is exposed only when need be. The impact of the IoT-based system in addressing water efficiency and an enhanced condition of the plants provided watering as opposed to being watered manually is thus proven in the graph.

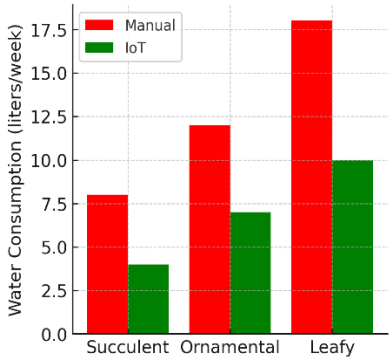


Fig. 5. Water consumption comparison between IoT-controlled and manual irrigation across different plant categories.

The IoT-based system made by employing less water compared to manual irrigation (V. Sharma et al.(2023)), was regular across all the types of plants as indicated in Fig. 5. In the case of succulents, almost half of the water was used as compared to ornamental plants, where a half-rate change was realized. The largest change was observed in leafy vegetables where the amount of water used has dropped as compared to 18 liters per week in the case of manual watering to only 10 liters with the IoT system. This shows that the system is very effective in timetabling irrigation schedules to certain irrigation requirements needed by individual plants and not merely using uniform water application. The IoT design (A. Singh et al. (2022)), will save precious water resources by averting needless irrigation without interfering with the well-being of plants. The findings also highlight the fact that the use of IoT in irrigation does not automatically digitise the activity but keeps it informative and accurate. Guess Work or visual sight often underlies manual irrigation which is liable to mistake. On the contrary, the system of the IoT is based on sensor data and maximum indicators, meaning that every plant will receive precisely the water quantity it will need. Such exactness lowers wastage, efficiency and directly facilitates sustainable water management. Lowered consumption of water has a particularly large payback in environmental benefit in areas where the problem of water shortage can persistently worry the population.(R. Jain et al.(2022)), (S.Sharma et al. (2023))

Along with the quantitative data, the conceptual observation of the quality of the plant growth was realized. IoT-controlled irrigation plants were always in excellent conditions with the root zones being in a favorable moisture level. The method of manually watering soil, leaving it irregular with wet and dry parts could easily lead growth retardation or stressed crops. Even though this point is not quantitative in its essence, it describes a significant practical benefit of applying IoT in agriculture and home gardening. In summary, the findings support the stated hypothesis that the suggested IoT-enabled watering system at plants is effective to eliminate the constraints that manual irrigation has. The system conserves water and focuses on better plant care since it ensures punishment of soil soil stays at the best moisture level and eliminate wastage of water. Moreover, the design is reproducible and scalable across different plants with different requirements making it an adaptable solution that fits both small size gardens and large farming requirements.

Conclusion

As considering the design of a very simple IoT-based plant watering system making use of soil moisture sensors and microcontroller to direct water supply in this paper. The primary goal was to lessen the speculation in hand watering and to make sure that every plant receives the affluent level of water. Johnson and Jordanuces observe and compare such that it is apparent that the system enables the soil to be moistened more consistently than the manual intervention of watering that tends to cause extreme highs and lows. The importance of this regular watering schedule is that it allows the plants to be in good health as well as avoid unnecessary consumption of water.

The convenience of the usage is another thing that emerged out of this work. The system will be connected to an IoT platform; as a result, the user is able to watch the state of the plants regardless of where he or she is and does not necessarily have to always be physically around. This renders the system not only viable in households, but also in case of small-scale nurseries and farms whereby it might not be possible to supervise the system regularly. On the whole, the article indicates that the integration of simple sensors and IoT can make irrigation issues a low-cost and efficient solution..

Future Scope

Even though the system works as expected for the present setup, there are many ways in which it can be improved in the future. One possible improvement is to add more sensors that can measure not only soil moisture but also temperature, humidity, and maybe even soil nutrients. This would allow the system to make better decisions and provide even more accurate watering schedules. Another idea is to connect the system with local weather data so that it can automatically skip irrigation if rain is predicted.

For long-term use in farms, the system can also be combined with solar panels so that it does not depend on grid electricity. This would make it more reliable in rural areas and reduce running costs. Another interesting direction would be to store the collected data and analyze it over time. If the system can learn patterns of water usage for different plants, it could automatically adjust thresholds for better results. With these kinds of improvements, the IoT watering system could grow into a more advanced tool that supports sustainable agriculture (A. Verma et al.(2023)), and helps in saving water on a larger scale.

There is much space in the way of additional development of the proposed plant watering system based on IoT. The simplest path to go is incorporation of more sensors like the humidity, temperature and pH sensors, which would then enable the system to make more/better informed watering decisions than what the soil moisture alone would achieve. This would most especially help on crops which are sensitive to various environmental factors. The final opportunity to contextualize the system is the introduction of renewable source energy, (R. Jain et al.(2022)), e.g., in the form of solar panels,. to turn the system into power grid-independent and adapted to rural or remote territories.

Through linking more than one node in a wireless sensor network, the project may also be extended to larger fields. This would enable the farmers to manage and oversee irrigation method on expansive agricultural lands through one set of dashboard. It could be further enhanced with building an integration with weather forecast(A. Singh et al.(2023)), services to make sure that plants will not be watered with the rainfall to decrease the tipping over of unnecessary water use. In case of first detail, resonance with urban applications, the system would be integrated with home automation platforms, thus ensuring that it integrates well with smart homes.

The system might be extended with mobile alerts and predictive models in the long run, indicating to the farmers or gardeners when this is the right time to water and when to apply fertilizer or take precautions against plant health. With a chance of commercialization, the project can make a contribution to sustainable agriculture as it will save on water, reduce costs, and increase productivity.

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